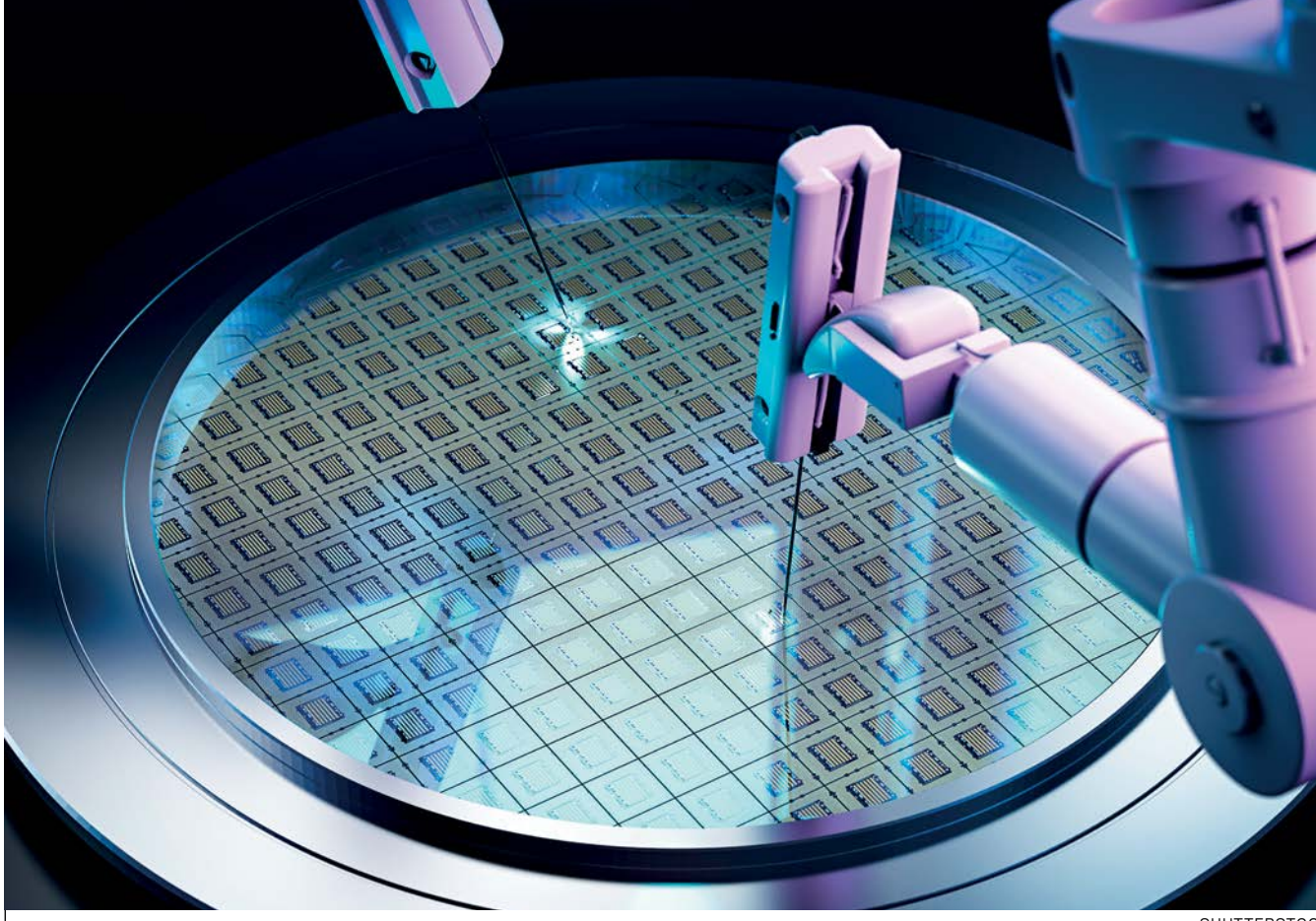


SHUTTERSTOCK

INDIA@100 | COMPUTING SEMICONDUCTORS

CONDUCTING THE CHIPS OPERA

SEMICONDUCTORS HAVE BECOME INTEGRAL TO OUR EXISTENCE, WHICH MAKES THEIR DOMESTIC MANUFACTURING VITAL FOR THE ECONOMIC INDEPENDENCE AND NATIONAL SECURITY OF INDIA AND OTHER COUNTRIES

An entire car ready on the assembly line but unable to ship, merely because the waiting period for a thumbnail-sized chip to run its collision avoidance system is running into months. The shortage of electronic chips during the peak of Covid-19 drove home a crucial point. That, today, semiconductors are embedded in nearly every device you can think of—from home appliances like televisions and refrigerators to smartphones and cars. But the fabrication of these chips is limited to just a few places worldwide, creating the risk of global supply chain disruptions during any Covid-like crisis. At present, Taiwan produces 90 per cent of the world's most advanced semiconductors. So, to protect themselves from disruptions, global corporations and governments are investing billions of dollars to boost domestic chip manufacturing.

WHY IT IS A GAME CHANGER

India is setting out to build a semiconductor manufacturing ecosystem, having announced an outlay of Rs 76,000 crore for a production-linked incentive (PLI) scheme in December 2021. The market for semi-